

Homework 1

Haoyang Sun

Abstract

This is my abstract!

1. QUESTION 1

1.a. *Proof*

Start by taking logs on both sides of the return expression:

$$\log R_{e,t+1} = \log(1 + \exp(\log D_{t+1} - \log P_{t+1})) + \log P_{t+1} - \log P_t \quad (1)$$

For simplicity, I denote $r_{e,t+1} = \log R_{e,t+1}$, $p_{t+1} = \log P_{t+1}$, $d_{t+1} = \log D_{t+1}$, and $dp_{t+1} = \log D_{t+1} - \log P_{t+1}$, then, by a Taylor expansion in dp_{t+1} (which is stationary with mean $\bar{dp}$), we get

$$r_{e,t+1} = \log(1 + \exp(\bar{dp})) + \frac{\exp(\bar{dp})}{1 + \exp(\bar{dp})} (dp_{t+1} - \bar{dp}) + p_{t+1} - p_t. \quad (2)$$

Denote $\kappa = 1/(1 + \exp(\bar{dp}))$, and $\kappa_0 = \log(1 + \exp(\bar{dp})) - \bar{dp} \exp(\bar{dp})/(1 + \exp(\bar{dp})) = -\log \kappa - (1 - \kappa) \log(1/\kappa - 1)$. The log-linearization becomes

$$r_{e,t+1} = \kappa_0 + (1 - \kappa) dp_{t+1} + p_{t+1} - p_t. \quad (3)$$

Moving p_t to the left-hand side and the other terms to the right-hand side, we get

$$\begin{aligned} p_t &= \kappa_0 + (1 - \kappa) d_{t+1} - r_{e,t+1} + \kappa p_{t+1} \\ &= \kappa_0 + (1 - \kappa) d_{t+1} - r_{e,t+1} + \kappa [\kappa_0 + (1 - \kappa) d_{t+2} - r_{e,t+2} + \kappa p_{t+2}] \\ &= \kappa_0 (1 + \kappa + \dots + \kappa^{H-1}) \\ &\quad + (1 - \kappa) (d_{t+1} + \kappa d_{t+2} + \dots + \kappa^{H-1} d_{t+H}) \\ &\quad - [r_{e,t+1} + \kappa r_{e,t+2} + \dots + \kappa^{H-1} r_{e,t+H}] \\ &\quad + \kappa^H p_{t+H} \end{aligned} \quad (4)$$

The log dividend-price ratio dp_t can then be written as

$$\begin{aligned}
dp_t &= -\kappa_0(1 + \kappa + \dots + \kappa^{H-1}) \\
&\quad + [r_{e,t+1} + \kappa r_{e,t+1} + \dots + \kappa^{H-1} r_{e,t+H}] \\
&\quad d_t - (1 - \kappa)(d_{t+1} + \kappa d_{t+2} + \dots + \kappa^{H-1} d_{t+H}) \\
&\quad - \kappa^H p_{t+H} \\
&= -\kappa_0(1 + \kappa + \dots + \kappa^{H-1}) \\
&\quad + [r_{e,t+1} + \kappa r_{e,t+1} + \dots + \kappa^{H-1} r_{e,t+H}] \\
&\quad d_t - d_{t+1} + \kappa(d_{t+1} - d_{t+2}) + \dots + \kappa^{H-1}(d_{t+H-1} - d_{t+H}) \\
&\quad + \kappa^H d_{t+H} - \kappa^H p_{t+H} \\
&= \kappa_0 \frac{\kappa^H - 1}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H dp_{t+H}
\end{aligned} \tag{5}$$

where $\Delta d_{t+1} := d_{t+1} - d_t$. Notice that the identity above holds ex post (state by state), so it also holds in conditional expectation at time t , giving

$$dp_t = \kappa_0 \frac{\kappa^H - 1}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} + \kappa^H \mathbb{E}_t dp_{t+H} \tag{6}$$

Letting $H \rightarrow \infty$ and imposing the no-bubble (transversality) condition

$$\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t dp_{t+H} = 0 \tag{7}$$

we obtain equation (1.3):

$$dp_t = \kappa_0 \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} \tag{8}$$

1.b.

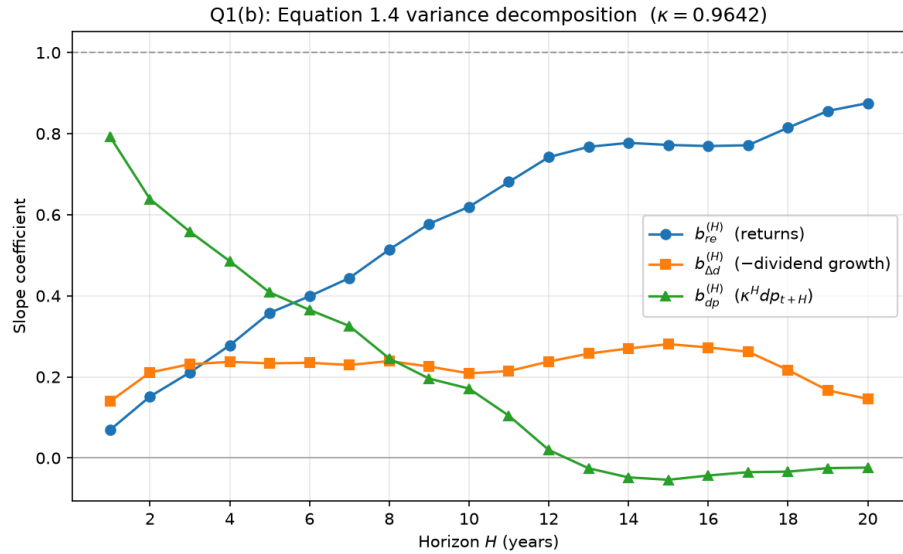


Figure 1: Equation 1.4 variance-decomposition slopes vs. horizon H (EQ Dataset, 1928–2021; overlapping monthly starts; $\kappa = 1/(1 + e^{\bar{d}^p}) = 0.9642$).

We can infer from the figure that, valuation ratio dp_t can predict log run return, and the volatility of price mainly comes from the volatility of discount rate.

1.c. VAR

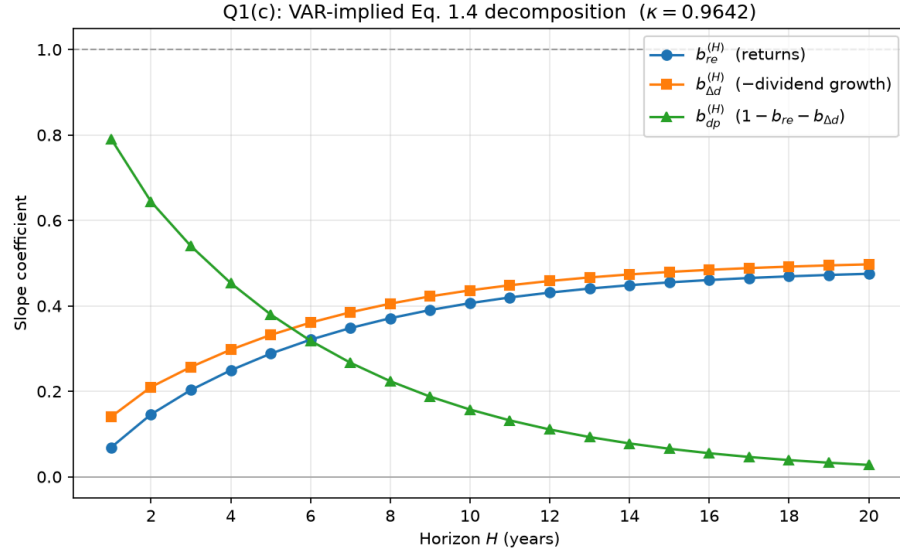


Figure 2: VAR-implied Equation 1.4 variance-decomposition terms vs. horizon H .

Overlapping-annual VAR(1) on $Z_t = [\Delta d_t, r_{e,t}, dp_t]'$ estimated by OLS (EQ Dataset, 1928–2021; $\kappa = 0.9642$, as in 1(b)). $b_{dp}^{(H)}$ is imposed as $1 - b_{re}^{(H)} - b_{\Delta d}^{(H)}$.

1.d.

1.e. Alternative log-linear present-value identity

Using the same trick,

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{D_{t+1}} \frac{D_{t+1}}{D_t} \frac{D_t}{P_t + D_t} \frac{P_t + D_t}{P_t} \quad (9)$$

, the log one-period return can be written as

$$r_{e,t+1} = -\log(1 - e^{-dy_{t+1}}) + \Delta d_{t+1} + \log(1 - e^{-dy_t}) + dy_t \quad (10)$$

where $dy_t := \log\left(1 + \frac{D_t}{P_t}\right)$ and $\log(1 - e^{-dy_t}) = \log \frac{D_t}{P_t + D_t}$.

Applying a Taylor expansion of $\log(1 - e^{-dy_t})$ in dy_t around its mean $\bar{dy}$, we have

$$\log(1 - e^{-dy_t}) = \log(1 - e^{-\bar{dy}}) + \frac{e^{-\bar{dy}}}{1 - e^{-\bar{dy}}} (dy_t - \bar{dy}) + O((dy_t - \bar{dy})^2) \quad (11)$$

Letting $\kappa = e^{-\bar{dy}}$ to simplify the expression,

$$\log(1 - e^{-dy_t}) = \log(1 - \kappa) + \frac{\kappa}{1 - \kappa} (dy_t - \bar{dy}) + O((dy_t - \bar{dy})^2) \quad (12)$$

Then the return is log-linearized as

$$r_{e,t+1} = -\frac{\kappa}{1 - \kappa} (dy_{t+1} - \bar{dy}) + \Delta d_{t+1} + \frac{\kappa}{1 - \kappa} (dy_t - \bar{dy}) + dy_t \quad (13)$$

and

$$\begin{aligned}
\frac{dy_t}{1-\kappa} &= r_{e,t+1} - \Delta d_{t+1} + \kappa \frac{dy_{t+1}}{1-\kappa} \\
&= r_{e,t+1} - \Delta d_{t+1} \\
&\quad + \kappa \left[r_{e,t+2} - \Delta d_{t+2} + \kappa \frac{dy_{t+2}}{1-\kappa} \right] \\
&= r_{e,t+1} - \Delta d_{t+1} + \kappa [r_{e,t+2} - \Delta d_{t+2}] \\
&\quad + \dots + \kappa^{H-1} [r_{e,t+H} - \Delta d_{t+H}] + \kappa^H \frac{dy_{t+H}}{1-\kappa} \\
&= \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H \frac{dy_{t+H}}{1-\kappa}
\end{aligned} \tag{14}$$

Thus, the log-linearization identity is

$$dy_t = (1-\kappa) \left(\sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H} \tag{15}$$

Taking the conditional expectation at time t , the identity also holds

$$dy_t = (1-\kappa) \left(\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} \right) + \kappa^H \mathbb{E}_t dy_{t+H} \tag{16}$$

Imposing the no-bubble assumption, the last term vanishes as $h \rightarrow \infty$,

$$\lim_{H \rightarrow \infty} \kappa^H E_t \log \left(\frac{P_t + D_t}{P_t} \right) \rightarrow 0 \tag{17}$$

2. QUESTION 2